



Smart World

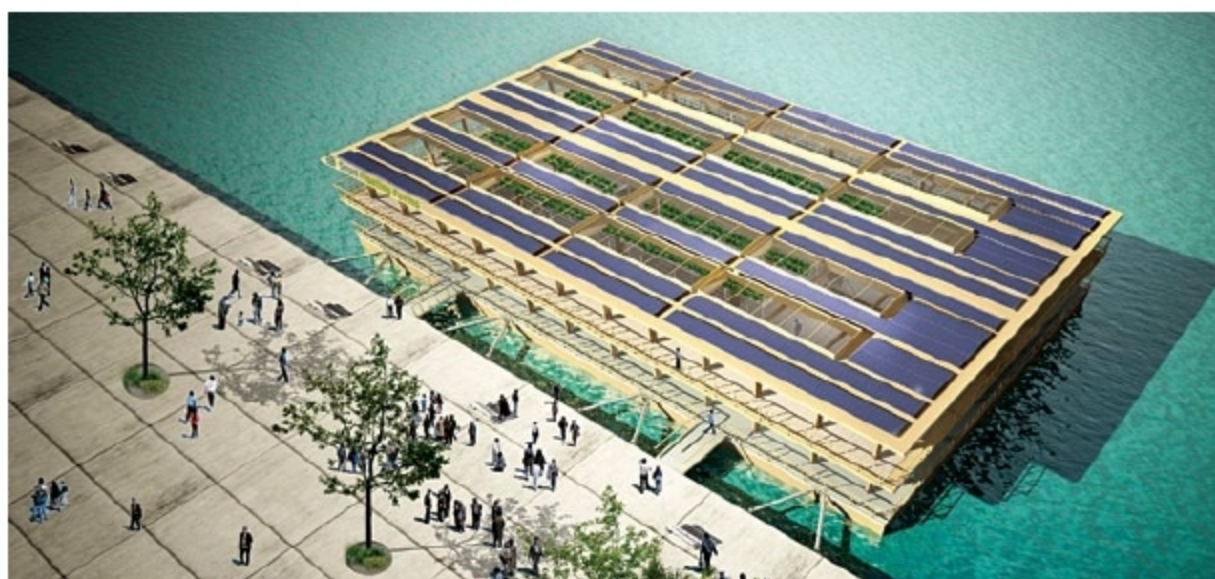
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1 FIGHTING WORLD HUNGER WITH SOLAR ENERGY

Although carbon-neutral initiatives are increasing, without effective implementation, the world target of reaching zero hunger by 2030 will not be possible



(Credit: <https://smartfloatingfarms.com>)

Technology has become an indispensable part of our lives, but we still find ourselves complaining about minor inconveniences every now and then. While for most of us imagining life without technology is difficult, there is a large population that is struggling to have access to even basic human needs. According to a 2019 report by Food and Agriculture Organization of the UN, more than 820 million people are still suffering from hunger globally and the number is continuously rising. One such example is Zimbabwe, where more than five million people are currently facing this problem.

Amidst dwindling natural resources, we need to consider viable solu-

tions to meet global demands of nutritional food for the growing population and improve the livelihood of people. What are the major reasons behind food scarcity, and how solar can make a significant impact in tackling the issue?

Unsustainable ways of farming involve deforestation and greenhouse gas emissions, which result in climate change. Unsuitable weather conditions, such as droughts and floods, continue to ruin crops. Proper utilisation of available lands and irrigation is the need of the hour to solve this issue. As solar panels become more cost-effective every day, these can not only aid in achieving this aim but also reduce the carbon footprint.

For instance, SunCulture makes affordable solar-powered machines for irrigation. The machines use solar panels to pump water from any source in the vicinity. The amount of water available per day depends on distance between the pump and source. A mobile app is used to design customised irrigation systems with inbuilt sprinklers for farmers, thereby giving a high yield while conserving water.

Many such companies as Golden State Clean Energy have been converting unused lands to solar farms. Electricity produced is used for powering various agricultural activities. Unoccupied space in these large patches of land can be further utilised for farming. An example is north-eastern Japan where farms have grown cloud-ear mushrooms below solar panels, since these mushrooms thrive in dark and humid conditions.

A smart floating farm is another alternative to land-based farming. The top layer in this farm has photovoltaic cells to generate electricity. Openings for the lower layer ensure that sunlight reaches the plants. Other renewable energy sources like wind turbines and systems for power generation from waves are also available. Besides a rainwater collector and a water recycling system is present to meet water demands. Plants are grown in a hydroponic system without soil, thereby